

Installing and Using Software on CARC Systems

Cesar Sul

Research Computing Associate



USC

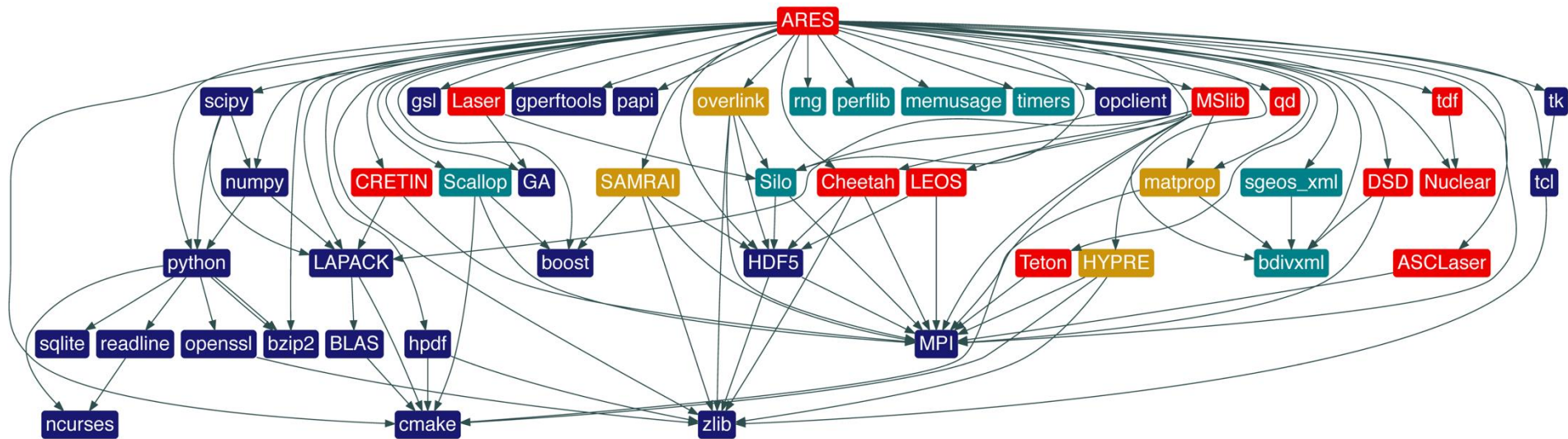
Advanced Research Computing
Enabling scientific breakthroughs at scale

Outline

- CARC managed software
 - Finding software
 - Using modules
 - Neat features
- Installing software
 - Precompiled binary
 - Conda
 - Python
 - Apptainer
 - R
 - Building Source Code

Software is complex!

- Dependency graph for ARES



<https://computing.llnl.gov/projects/spack-hpc-package-manager>

What are software modules?

- Modules present installed software to users
- Set environment variables
 - PATH
 - PKG_CONFIG_PATH
 - LD_LIBRARY_PATH
 - <SOFTWARE>_ROOT
- Show how package was built
- Show where package was installed to
- Prevent loading incompatible software
- Written in Lua

What are software modules?

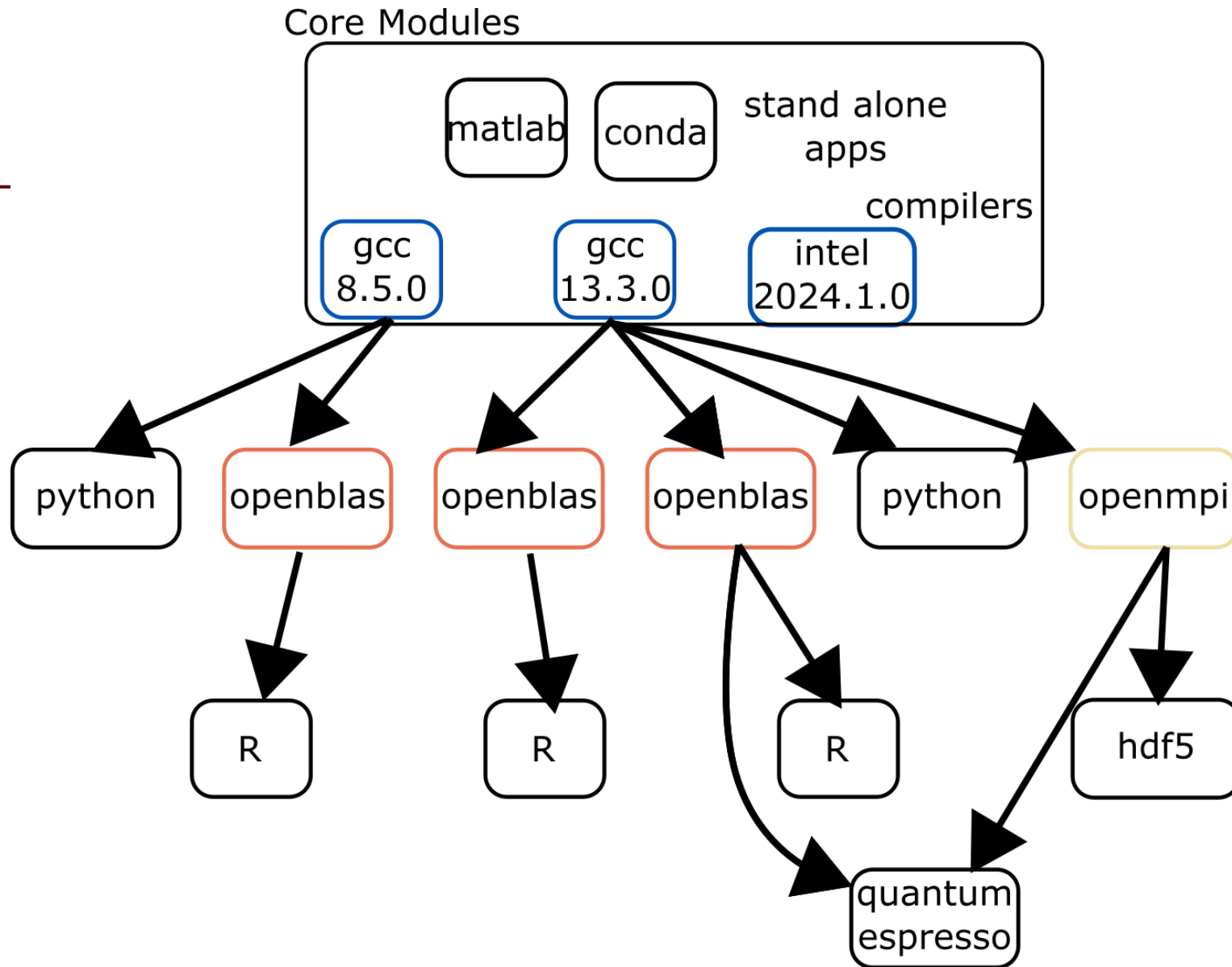
- There are 6 kinds of modules
 - Utilities
 - git, screen, tmux, wget,
 - Compiler (gcc, intel)
 - BLAS
 - openblas, mkl, amd-blis
 - Math Library
 - MPI
 - OpenMPI, mvapich2, IntelMPI
 - Multi node communication
 - Application
 - most common, bamtools, cmake, libpng...
 - Other
 - Matlab, Mathematica, julia,

What are software modules?

- By default you have the recommended "usc" module loaded

```
$ module list
Currently Loaded Modules:
  1) gcc/13.3.0          3) openblas/0.3.28    5) usc/13.3.0
  2) python/3.11.9      4) openmpi/5.0.5
```

- You can check what's available with [module avail](#)
- Depending on active modules, you will see different results from [module avail](#)



Example module avail listing

\$ module avail

[/spack/apps/lmod/linux-centos7-x86_64/openmpi/4.0.2-ipm3dnv/openblas/0.3.8-2no6mfz/gcc/8.3.0](#)

cantera/2.4

Applications built with gcc 8.3.0 compiler
AND openmpi 4.0.2 AND openblas 0.3.8

openblas-openmpi

hypr/2.18

[/spack/apps/lmod/linux-centos7-x86_64/openmpi/4.0.2-ipm3dnv/gcc/8.3.0](#)

hdf5/1.10.6-openmpi

Applications built with gcc 8.3.0 compiler AND openmpi 4.0.2

openmpi/2.29-openmpi

hmmer/3.3-openmpi

matio/1.5.13-openmpi

parmetis/4.0.3-openmpi

sundials/5.1.0-openmpi (D)

[/spack/apps/lmod/linux-centos7-x86_64/openblas/0.3.8-2no6mfz/gcc/8.3.0](#)

r/3.4.4

Applications built with gcc 8.3.0 compiler AND openblas 0.3.8

adapterremoval/2.3.1

kbproto/1.0.7

pdt/3.25.1

anaconda3/2020.02

Applications built with gcc 8.3.0 compiler

er/2.173

argtable/2.13

ale/1.05

at-spi2-atk/2.26.2

lcms/2.9

perl-extutils-config/0.008

How to use modules

- Use **module avail** to see what's available
- Use **module load** to load the module

```
$ which python  
/usr/bin/python
```

```
$ module load python
```

```
$ which python  
/apps/spack/2406/apps/linux-rocky8-x86_64_v3/gcc-  
13.3.0/python-3.11.9-x74mtjf/bin/python
```

- Some modules 'unlock' more modules
 - Compiler
 - MPI
 - BLAS

Finding modules

- Use [module spider](#) to search for software that's not available
 - Might be hidden due to prerequisites

```
$ module spider r/4.4.2
```

```
-----  
r: r/4.4.2  
-----
```

You will need to load all module(s) on any one of the lines below before the "r/4.4.2" module is available to load.

```
gcc/13.3.0
```

Saving sets of modules

- If you find yourself loading a set of modules frequently

<code>module save</code>	Save current modules to default collection
<code>module save <name></code>	Save current modules as <code><name></code> collection
<code>module restore</code>	Load modules in default collection
<code>module restore <name></code>	Load modules in <code><name></code> collection
<code>module describe <name></code>	Show which modules are in <code><name></code> collection
<code>module savelist</code>	Show names of all collections

Creating your own modules

- Check our Discourse page for more details
- [How to: Create our own modules](https://hpc-discourse.usc.edu/how-to-create-your-own-modules/149)

The screenshot shows a web browser displaying a Discourse forum post. The browser's address bar shows the URL `https://hpc-discourse.usc.edu/how-to-create-your-own-modules/149`. The page header includes the USC logo and the text 'Advanced Research Computing' and 'Enabling scientific breakthroughs at scale'. A blue banner at the top of the post area contains a message about bootstrap mode. The post title is 'How to: Create your own modules' with a pencil icon. Below the title, it says 'HPC Cluster: Discovery' and 'Software Modules'. The post is by user 'csul' and was posted '4d' (4 days) ago. The post content begins with 'After building and installing your own software, you might find it convenient to add it to your personal module tree.' and lists three basic steps: 1. Create module directory tree, 2. Add module directory tree to \$MODULEPATH, and 3. Create module file from template. A diagram illustrates the directory structure: a yellow box labeled '\$MODULE_ROOT' is at the top. Below it are two red boxes labeled 'software1/' and 'software2/'. Under 'software1/' is a blue box labeled 'ver-1.lua'. Under 'software2/' are two blue boxes labeled 'ver-1.lua' and 'ver-2.lua'. The post continues with text explaining the structure and the need for module files. The bottom of the screenshot shows the start of the second step: '2. Add module directory tree to \$MODULEPATH'.

To make launching your new site easier, you are in bootstrap mode. All new users will be granted trust level 1 and have daily email summary emails enabled. This will be automatically turned off when 50 users have joined.

How to: Create your own modules

HPC Cluster: Discovery Software Modules

csul 4d

After building and installing your own software, you might find it convenient to add it to your personal module tree.

There are three basic steps:

1. Create module directory tree
2. Add module directory tree to \$MODULEPATH
3. Create module file from template

Read below for more details on each step.

1. Create module directory tree

You'll need to set up a directory tree where all your module files will be saved to. You can place the root of the directory tree anywhere but we'll use \$MODULE_ROOT for this document.

You can look at /spack/apps/mod/linux-centos7-x86_64/ to get an idea of the structure we use at CARC.

Since CARC manages so many applications we have our module tree separated by compiler used, MPI library, and BLAS library. This is probably excessive for most users so you might want a structure like:

```
graph TD
    A["$MODULE_ROOT"] --> B["software1/"]
    A --> C["software2/"]
    B --> D["ver-1.lua"]
    C --> E["ver-1.lua"]
    C --> F["ver-2.lua"]
```

In this example there are 2 directories under \$MODULE_ROOT, software1 and software2. Each of these software directories has at least 1 module file written in Lua (don't worry, there's a template you can use below).

2. Add module directory tree to \$MODULEPATH

After creating the directory tree you can add it to your \$MODULEPATH by running the command

Installing Your Own Software



USC

Advanced Research Computing
Enabling scientific breakthroughs at scale

Installing software

- Installing software can be quick and painless
 - With precompiled binaries for your specific operating system, it can be as easy as unzipping a file
- ... Or neither quick nor painless
 - If you have to compile the software yourself using compilers, linkers, Makefiles, external libraries, etc.
- (or even worse...)
 - If it's from an academic lab from 1999 and requires old versions of multiple libraries which have multiple dependencies!

Installing software

- Generally speaking, software can be installed globally or locally
 - On your laptop, you are the system administrator
 - On HPC, you are not the system administrator
- Globally means system-wide
 - Software is installed to system locations like `/usr/bin` or `/usr/local`
 - Global installs require root privileges
- System-wide installations will not work on HPC
 - Only systems administrators have root privileges on HPC
 - E.g., “yum install” and “apt install” will not work



Installing software

- CARC users must perform local, or “user”, installs
 - Software installed to 'local' folders
 - [/project2/<pi_id>/software](#)
 - Requires write privileges, which you have in your own directories
 - Software will be accessible by you, even on compute nodes
- It is not always obvious how to perform a user install
 - Depends on software
 - You may have to check documentation

Precompiled binary

- Simplest case
- Just download and extract
- Not always available

```
$ cd /project2/ttroj_412/software
```

```
#Copy tarball
```

```
$ wget https://example.com/sample.tar.gz
```

```
#Extract files
```

```
$ tar xvf sample.tar.gz
```

```
#Set your environment (adds a new location to your path)
```

```
$ export PATH=/project2/ttroj_412/software/sample/bin:${PATH}
```

```
#Test installation
```

```
$ binary_name
```

Conda Environments



USC

Advanced Research Computing
Enabling scientific breakthroughs at scale

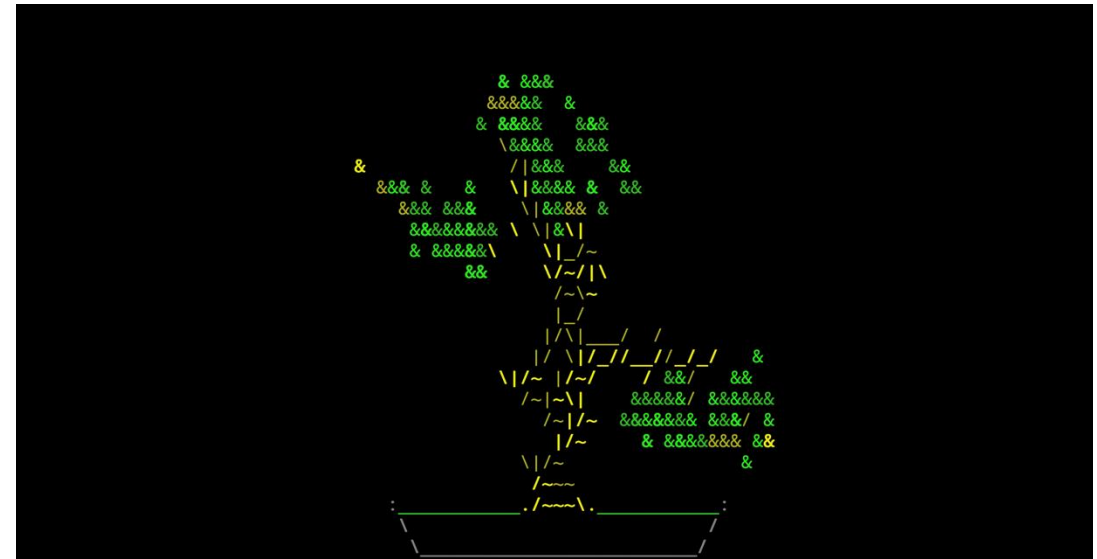
Conda Environments

- Some developers package their software as conda environments
- Create a collection of software packages
- Dependencies are managed for you
- `mamba env create -f environment.yml`
- Full documentation [here](#)

```
name: mustache
channels:
  - conda-forge
  - defaults
dependencies:
  - pip
  - h5py
  - hdf5
  - numpy
  - python=3.8
  - pip:
    - cooler
    - hic-straw
    - pandas
    - scipy
    - statsmodels
```

Our own environment

- We can also create our own custom environments
- Let's use "cbonsai" example
- Prints ascii art of bonsai tree



Our own environment

- Set up conda

```
$ module load conda
```

```
$ mamba init bash
```

- Create and name environment

```
$ conda create -n cbonsai
```

```
# Enter environment
```

```
$ conda activate cbonsai
```

- Install packages

```
$ mamba install cbonsai
```

```
$ mamba install cbonsai
```

```
.  
.br/.
```

Package	Version	Build	Channel	Size
---------	---------	-------	---------	------

Install:

+ _libgcc_mutex	0.1	conda_forge	conda-forge	3kB
+ libgomp	14.2.0	h77fa898_1	conda-forge	461kB
+ _openmp_mutex	4.5	2_gnu	conda-forge	24kB
+ libgcc	14.2.0	h77fa898_1	conda-forge	849kB
+ ncurses	6.5	h2d0b736_3	conda-forge	892kB
+ cbonsai	1.3.1	haa1a288_0	conda-forge	30kB

Summary:

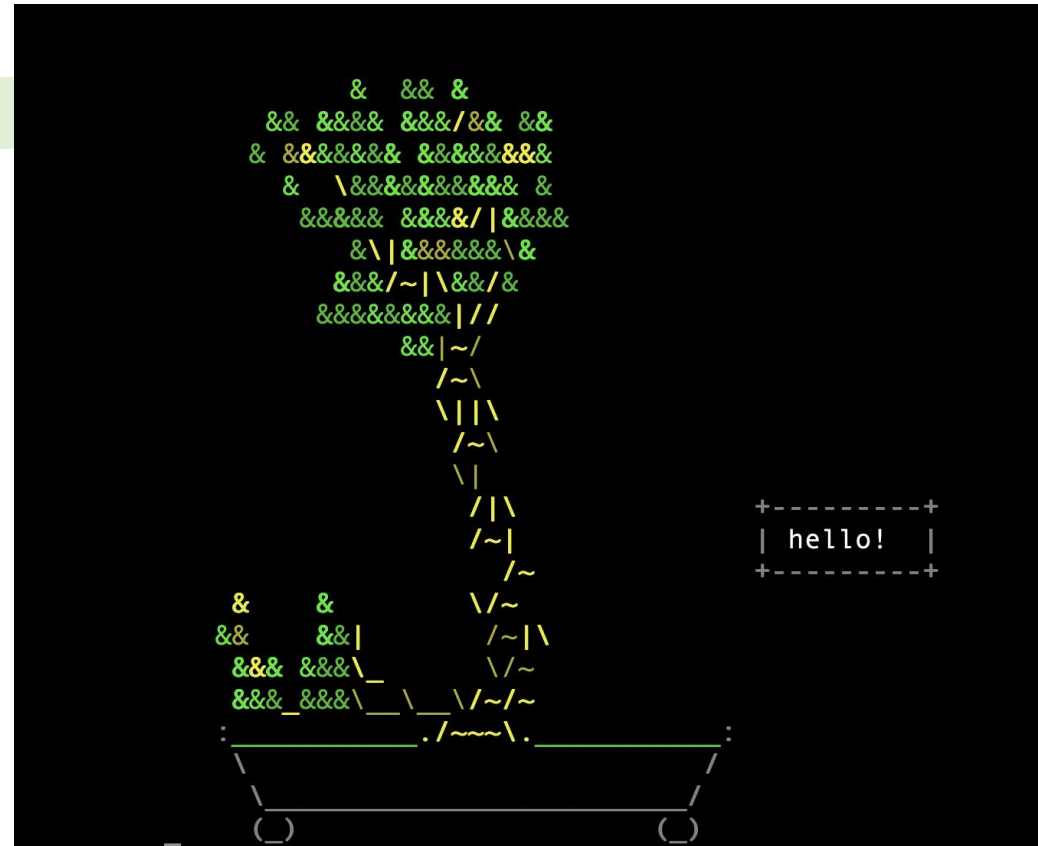
Install: 6 packages

Total download: 2MB

Our own environment

- Run cbonsai

```
$ cbonsai -m hello! -p
```



Installing Python Packages



USC

Advanced Research Computing
Enabling scientific breakthroughs at scale

Installing Python packages

- Don't forget to load the version of python you want to use

```
$ module load python/3.7.6
```

- To check what packages are available use the command

```
$ pip list
```

- Install package (bash shell)

```
$ pip install <package_name>
```

- Sometimes you'll need to install the latest version of a package that is already installed

```
$ pip freeze  
$ pip install <package_name> --upgrade
```

Dependencies for Python packages

- Some packages are Python wrappers for C/C++ libraries
- The installer needs to know where these libraries are
- The [h5py](#) package is one example

```
$ HDF5_DIR=/path/to/hdf5  
$ HDF5_VERSION=X.Y.Z  
$ CC="mpicc"  
$ pip install h5py --user
```

- You might have to download the package tarball and edit some files like setup.py

Apptainer/Containers



USC

Advanced Research Computing
Enabling scientific breakthroughs at scale

Apptainer



-
- Formerly Singularity
 - For difficult installations
 - Apptainer provides packaged "computing environments"
 - Works best with complex dependency chains
 - Compatible with Docker

Apptainer

Example: [lolcow](#) (date | cowsay | lolcat)

```
#Download container image
$ apptainer pull docker://ghcr.io/apptainer/lolcow

#Test
$ apptainer run lolcow_latest.sif
```

See this page for more ways to interact with a container:
https://apptainer.org/docs/user/latest/quick_start.html#interacting-with-images

```
[ttroj@discovery playground]$ apptainer run lolcow_latest.sif
```

```
< Fri Feb 7 09:26:29 PST 2025 >
```

```
  \  ^  ^
   \ (oo)\_____.
    (__)\       )\/\
       ||----w |
       ||     ||
```

Apptainer

```
Bootstrap: library
From: ubuntu:16.04

%post
  apt-get -y update
  apt-get -y install fortune cowsay lolcat

%environment
  export LC_ALL=C
  export PATH=/usr/games:$PATH

%runscript
  fortune | cowsay | lolcat

%labels
  Author GodloveD
```

```
[ttroj@discovery playground]$ apptainer run lolcow_latest.sif
```

```
< Fri Feb 7 09:26:29 PST 2025 >
```

```
  \  ^  ^
  \ (oo)\____
  ( _)\  )\
    ||----w |
    ||     ||
```

Installing R Packages



USC

Advanced Research Computing
Enabling scientific breakthroughs at scale

R on CARC

- Containers preferred
- R modules are actually containers
- You can create your own R containers in Apptainer
- Installed packages are available in container

Installing R packags

- Source the version of R you want to use and start R

```
$ module load r  
$ R
```

- Install package syntax (you may have to specify a path)

```
> install.packages('<package_name>')  
> install.packages('<package_name>', lib="/path/to/packages")
```

- Then load the library when you want to use

```
> library('<package_name>')  
> library('<package_name>', lib.loc="/path/to/packages")
```

Building Source Code

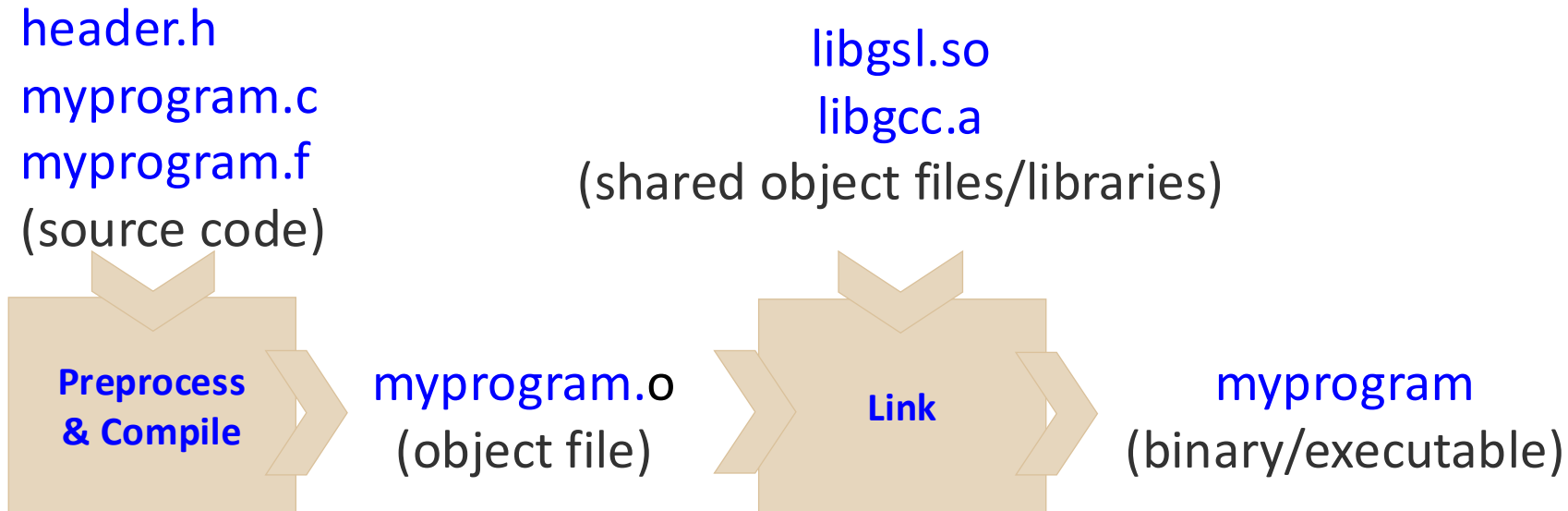


USC

Advanced Research Computing
Enabling scientific breakthroughs at scale

Compiling to source code

- C/C++ and Fortran programs are compiled and assembled



Compilers

- A typical compile command for C code

```
$ gcc ${CCFLAGS} source.c ${CPPFLAGS} ${LDFLAGS} -o myprogram
```

- Where these environment variables were pre-defined

```
$ CCFLAGS='-Wall -O3'
```

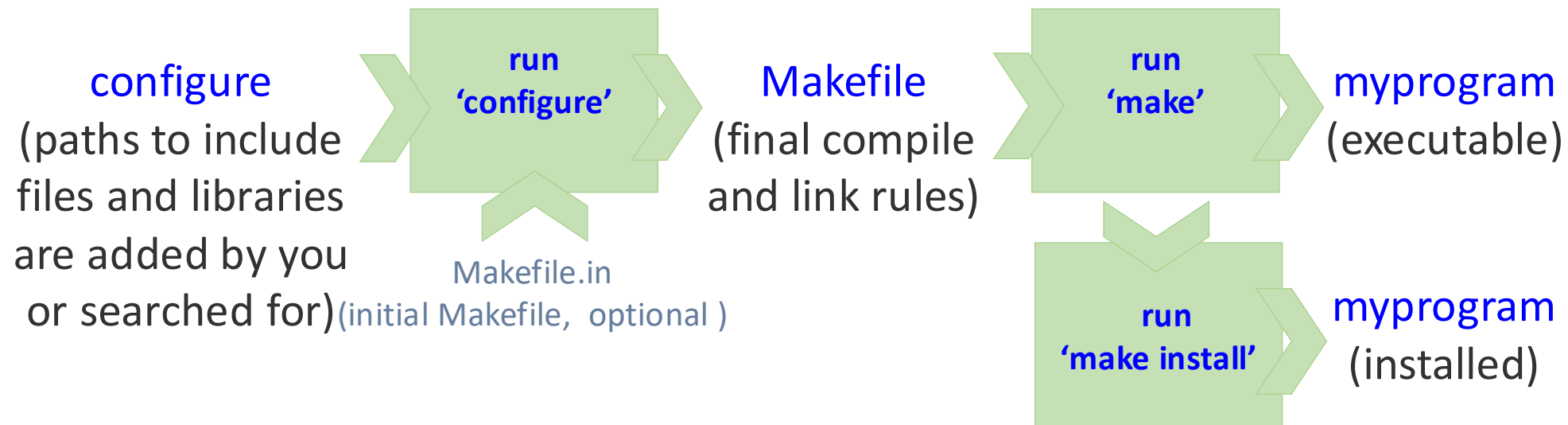
```
$ CPPFLAGS='-I/path/to/include'
```

```
$ LDFLAGS='-L/path/to/lib -lgsl -lgslcblas -lm'
```

CCFLAGS	Flags to pass the C compiler
CPPFLAGS	Where the C preprocessor can find include (.h) files
LDFLAGS	Which libraries (.so, .a files) to use and where the linker can find them

With configure/make

- Manually typing compile and link commands is not feasible
- Software build utilities like autotools, cmake handle this



Building software with modules

- Most modules modify `$PKG_CONFIG_PATH`
- Many installer scripts check here for prerequisite software

```
$ module load ncurses
$ ./configure <options>
...
checking curses.h usability... yes
checking curses.h presence... yes
checking for curses.h... yes
checking ncurses.h usability... yes
checking ncurses.h presence... yes
checking for ncurses.h... yes
...
```

- If we're lucky it's that easy

Building software with modules

- If unlucky, specify with script options

```
module load ncurses
./configure --ncurses-root=${NCURSES_ROOT} <other options>
...
checking curses.h usability... yes
checking curses.h presence... yes
checking for curses.h... yes
checking ncurses.h usability... yes
checking ncurses.h presence... yes
checking for ncurses.h... yes
...
```

- Our modules set environment variable `<software>_ROOT` for just this occasion

Building software with modules

- If very unlucky, modify Makefile

```
...  
override LDFLAGS += -L./nicksrc -L$(GSL_ROOT)/lib  
-L$(OPENBLAS_ROOT)/lib  
  
override CFLAGS += -c -g -p -Wimplicit -I./ -I./nicksrc  
-I$(GSL_ROOT)/include -I$(OPENBLAS_ROOT)/include  
...  
  
$ module load ncurses  
$ make
```


Compiling source code



Example: vim

#Download tarball

```
$ git clone https://github.com/vim/vim.git
```

```
$ cd vim
```

#Run configure script

```
$ ./configure --prefix=/project2/<pi_id>/<username>/vim [other options]
```

#Run makefile

```
$ make
```

```
$ make install
```

Compiling source code

- Configure script can't find libtinfo.so (provided by ncurses)

```
checking --with-tlib argument... empty: automatic terminal library selection
checking for tgetent in -ltinfo... no
checking for tgetent in -lncurses... no
checking for tgetent in -ltermcap... no
checking for tgetent in -ltermcap... no
checking for tgetent in -lncurses... no
no terminal library found
checking for tgetent()... configure: error: NOT FOUND!
```

Compiling source code

- In this case, pkg-config is not used
- Manually override with pkg-config, \$LDFlags, and \$CXXFlags

```
$ pkg-config --cflags-only-I ncurses
```

```
-D_GNU_SOURCE -I/spack/apps/linux-centos7-x86_64/gcc-8.3.0/ncurses-6.1-  
akiyo4qrgzlxw3hggkc42nvv7hz2evj/include
```

```
$ pkg-config --libs-only-L ncurses
```

```
-L/spack/apps/linux-centos7-x86_64/gcc-8.3.0/ncurses-6.1-akiyo4qrgzlxw3hggkc42nvv7hz2evj/lib
```

Compiling source code



Example: vim

```
#Download tarball
$ git clone https://github.com/vim/vim.git
$ cd vim

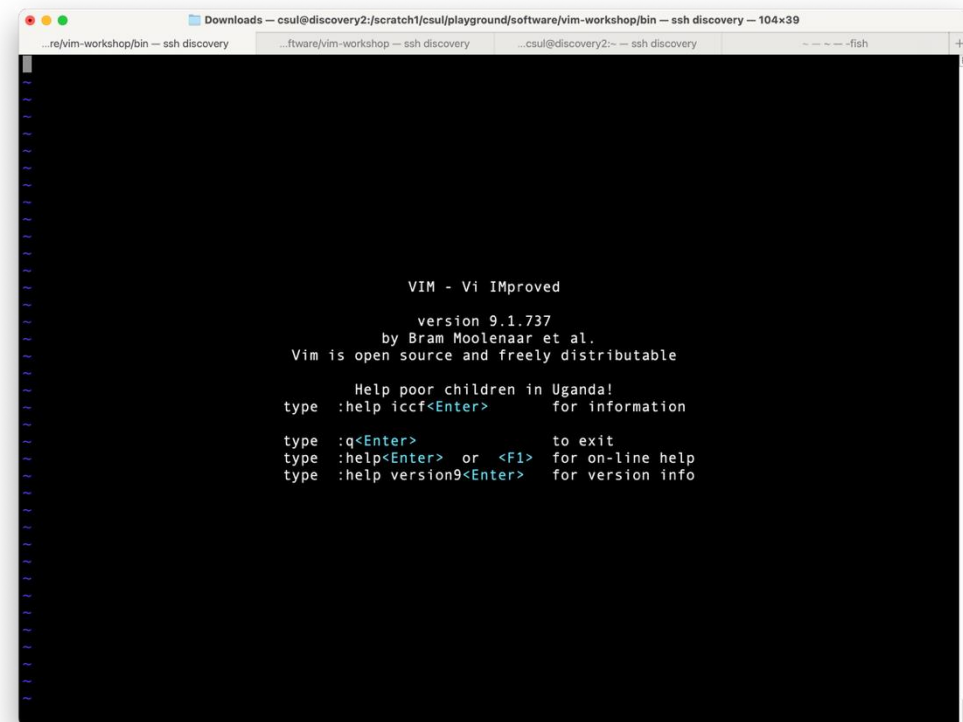
#Run configure script
$ CPPFLAGS=$(pkg-config --cflags ncurses) \
  LDFLAGS=$(pkg-config --libs-only-L ncurses) \
  ./configure \
  --prefix=/project2/<pi_id>/<username>/vim \
  --with-tlib=tinfo

#Run makefile
$ make
$ make install
```

Compiling source code

Test installation

```
$ ./vim # :q! to quit
```



The screenshot shows a terminal window with a dark background and light-colored text. The text is the Vim startup screen, which includes the version number 9.1.737, the author Bram Moolenaar, and a list of help commands. The terminal window has a title bar with the text "Downloads - csul@discovery2:/scratch1/csul/playground/software/vim-workshop/bin - ssh discovery - 104x39".

```
VIM - Vi IMproved
          version 9.1.737
    by Bram Moolenaar et al.
Vim is open source and freely distributable

  Help poor children in Uganda!
type  :help iccf<Enter>      for information

type  :q<Enter>              to exit
type  :help<Enter> or <F1>   for on-line help
type  :help version9<Enter> for version info
```

Getting Help

- Request assistance
 - Email carc-support@usc.edu
 - Office Hours (drop-in)
 - Every Tuesday@2:30pm (Zoom)
- Learn more!
 - Visit carc.usc.edu
 - Request a consultation (anytime)
 - Attend a Workshop (when scheduled)
 - [Visit our Discourse page!](#)



Thank you for attending!

Questions?

carc-support@USC.EDU



USC

Advanced Research Computing
Enabling scientific breakthroughs at scale